Polynomials — problem 10.69

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The polynomial $ax^2 + bx + c$ with $a > 0$ has real zeros x_1, x_2. Show that

$$|x_i| \leq 1,\ (i = 1, 2) \Leftrightarrow a + b + c \geq 0,\ a - b + c \geq 0,\ a - c \geq 0.$$